OpenOrbitLink: An Open-Source Framework for Delay-Tolerant Satellite Messaging with Elevation-Aware Rate Adaptation over LoRa LEO Links

Ashutosh Kumar
Independent Researcher
India
`github.com/Ashutosh0x/OpenOrbitLink`

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Abstract

We present **OpenOrbitLink**, an open-source framework for delay-tolerant satellite messaging over Low Earth Orbit (LEO) IoT constellations using LoRa modulation on ISM bands. The framework comprises a backend server (FastAPI), ground station driver (Raspberry Pi + SX1276), and Android client (Jetpack Compose). We propose an **elevation-aware rate adaptation algorithm** that selects spreading factors (SF7–SF12) based on real-time link budget computation during satellite passes, providing up to **5,469 bps at zenith** versus 293 bps with fixed SF12. Numerical link budget analysis at 868 MHz over a 550 km LEO path shows positive margin (+2.1 dB at 30° elevation with SF12), though we identify significant unaccounted losses (polarization mismatch, implementation loss) that reduce this margin in practice. We additionally implement LR-FHSS packet fragmentation and hopping sequence generation for future capacity scaling. The system uses BPv7 (RFC 9171) for store-and-forward relay and AES-256-GCM for end-to-end encryption. **No over-the-air satellite contacts have been achieved**; all results are numerical analysis based on datasheet parameters. The complete source code is available under MIT license.

Keywords: Satellite IoT, LoRa, LR-FHSS, Delay-Tolerant Networking, Adaptive Data Rate, LEO, Open-Source

1 Introduction

The integration of LoRa modulation with LEO satellite constellations has emerged as a promising approach for global IoT connectivity, particularly in areas lacking terrestrial infrastructure. Yu *et al.* [1] provide a comprehensive framework for LoRa-based LEO satellite IoT, identifying feasibility through numerical simulations with practical system parameters. Their work, along with error performance characterization by the same group [2], establishes the theoretical foundation for direct-to-satellite LoRa links.

Commercial satellite IoT services—including Lacuna Space, Swarm/SpaceX, and the announced Jio LEO constellation—operate on licensed spectrum with proprietary protocols. Meanwhile, open-source ground station networks such as TinyGS [13] and SatNOGS [14] have demonstrated successful LoRa satellite reception, and the FOSSASAT constellation [15] provides open-protocol IoT satellites on 868 MHz ISM.

OpenOrbitLink contributes an integrated open-source *framework* (not a deployed operational system) spanning:

1. An **elevation-aware rate adaptation algorithm** for LoRa spreading factor selection during LEO passes.
2. An **LR-FHSS packet fragmentation engine** with hopping sequence generation.
3. A **complete software stack** (backend + ground station driver + Android client) released under MIT license.

4. **Numerical link budget analysis** identifying both opportunities and challenges for 868 MHz ground-to-LEO uplink.

We explicitly note that this work is *framework-level*: no over-the-air satellite contacts have been performed. All quantitative results derive from link budget calculations using datasheet parameters.

2 Related Work

2.1 LoRa-to-LEO Satellite Links

The feasibility of LoRa direct-to-satellite (DtS) communication has been established by multiple groups. Doroshkin *et al.* [3] demonstrated experimental LoRa reception from a LEO cubesat, measuring RSSI and SNR across pass geometries. Fernandez *et al.* [4] conducted systematic link budget analysis for LoRa-to-LEO paths, identifying minimum elevation angles for each SF. The comprehensive survey by Fraire *et al.* [5] covers the LoRa-to-LEO landscape including channel modeling, MAC protocols, and constellation design.

2.2 Adaptive Data Rate for Satellite LoRa

Standard LoRaWAN ADR [9] was designed for static terrestrial deployments and performs poorly with rapidly changing satellite channels. Colavolpe *et al.* [6] proposed elevation-dependent parameter selection for LEO links. Our algorithm extends this approach with a two-phase selection incorporating SNR history.

2.3 LR-FHSS for Satellite IoT

Semtech's LR-FHSS specification [8] enables frequency-hopping transmission on SX126x devices, with demodulation requiring gateway-class receivers. Boquet *et al.* [7] analyzed LR-FHSS capacity for satellite IoT. We note that SX1276 hardware (used in our ground station) does *not* support LR-FHSS; our implementation is a software simulation for capacity analysis only.

2.4 DTN and Bundle Protocol

The Bundle Protocol has evolved from the experimental BPv6 (RFC 5050 [11], now obsolete) to the Standards-Track BPv7 (RFC 9171 [10]), with security provided by BPSec (RFC 9172 [12]). Open-source implementations include ION [16], dtn7-go [17], and μD3TN [18]. Our framework targets BPv7 compliance.

2.5 Open-Source Ground Stations

TinyGS [13] operates a worldwide network of LoRa ground stations that receive satellite telemetry. SatNOGS [14] provides open-source satellite ground station hardware and scheduling. Our ground station design builds on similar commodity hardware (RPi + SX1276) but adds a Python driver with adaptive SF selection.

2.6 Comparison with This Work

Table 1 positions OpenOrbitLink against prior systems.

Table 1: Comparison with Related Systems

System	OTA Tests	Adaptive SF	LR-FHSS	DTN	E2E Crypto	Mobile App	Open
TinyGS [13]	✓	–	–	–	–	–	✓
SatNOGS [14]	✓	–	–	–	–	–	✓
FOSSASAT [15]	✓	–	–	–	–	–	✓
Lacuna Space	✓	–	✓	–	–	–	–
Fraire *et al.* [5]	Partial	–	Discussed	✓	–	–	–
Yu *et al.* [1]	Numerical	–	Discussed	–	–	–	–
OpenOrbitLink	–	✓	✓*	✓	✓	✓	✓

*Software simulation only; SX1276 hardware does not support LR-FHSS.

3 System Architecture

3.1 Overview

OpenOrbitLink follows a four-tier architecture (Fig. 3.1):

$$\text{Phone} \xrightarrow{\text{HTTP}} \text{Backend} \xrightarrow{\text{gRPC}} \text{Ground Station} \xrightarrow{\text{868 MHz}} \text{Satellite} \tag{1}$$

Four-tier system architecture. The RF link (dashed red) has not been validated over the air.

3.2 Backend Server

The backend uses FastAPI with SQLAlchemy (async PostgreSQL) and exposes **27 REST endpoints** across six routers: authentication (JWT + invite codes), message queuing with ISM duty-cycle tracking, satellite radar (SGP4 + WebSocket at 1 Hz), adaptive modem profile selection, LR-FHSS capacity simulation, turbo compression, and network speed testing.

3.3 Ground Station

The ground station runs on a Raspberry Pi Zero 2W ($21.50) with an RA-02 SX1276 module ($7.15) and a quarter-wave monopole antenna ($2.38), totaling **$31.03**. The SX1276 driver supports hardware (SPI), simulation, and serial (RYLR896) modes. Default configuration: 868.000 MHz, SF12, BW125 kHz, CR 4/5, +14 dBm.

3.4 Android Client

The mobile client (Kotlin/Jetpack Compose) provides: (1) polar radar display with SGP4-based satellite plotting at 1 Hz, (2) throughput dashboard with adaptive modem visualization, (3) encrypted messaging with Room-based offline queue, and (4) a five-layer transport fallback (WiFi → Carrier NTN → LoRa → APRS → Local Queue).

3.5 Transport Fallback Chain

Figure 1 shows the automatic transport selection logic.

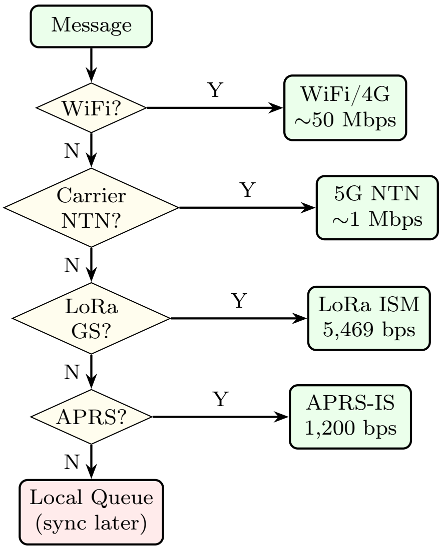

Figure 1: Five-layer transport fallback chain. The orchestrator selects the highest-priority available path.

3.6 Message Processing Pipeline

Figure 2 illustrates the end-to-end message flow from user input to RF transmission.

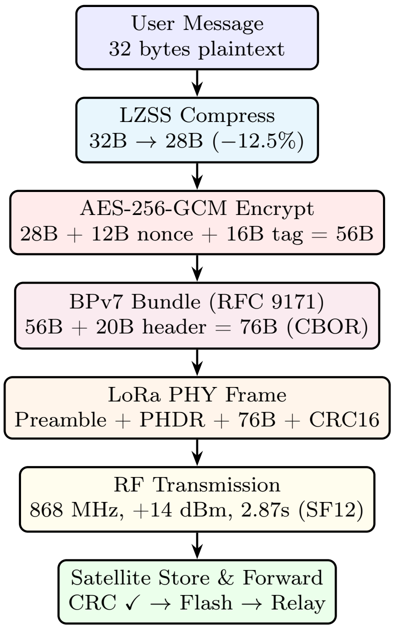

Figure 2: Message processing pipeline: user input through compression, encryption, BPv7 encapsulation, LoRa framing, and RF transmission.

4 Elevation-Aware Rate Adaptation

4.1 Motivation

Standard satellite LoRa deployments use fixed SF12 for maximum sensitivity (−137 dBm). However, during overhead passes the slant range decreases from ∼2,200 km (horizon) to ∼550 km (zenith), improving path loss by up to 12 dB. This excess margin can be traded for higher data rates by selecting lower spreading factors.

4.2 Link Budget Model

The received signal power at the satellite is:

$$P_{rx} = P_{tx} + G_{tx} - L_{total} + G_{rx} \tag{2}$$

The total path loss L_{total} includes:

$$L_{total} = L_{FSPL} + L_{atm} + L_{pol} + L_{impl} + L_{scint} \tag{3}$$

where the individual terms are:
Free-Space Path Loss:

$$L_{FSPL} = 20\log_{10}(d) + 20\log_{10}(f) - 147.55 \text{ [dB]} \tag{4}$$

Atmospheric Loss (ITU-R P.676 [21]):

$$L_{atm} = \frac{0.05}{\sin(\theta_{el})} \text{ [dB at 868 MHz]} \tag{5}$$

Polarization Mismatch Loss: A linear ground-station antenna and a tumbling satellite (random polarization) incur an average 3 dB loss, with deep fades exceeding 10 dB [4].

$$L_{pol} = 3.0 \text{ dB (average), up to 20 dB (fading)} \tag{6}$$

Implementation Loss: Accounts for cable/connector loss, LNA noise figure above datasheet, non-ideal filtering, and body/environmental shadowing:

$$L_{impl} = 2.0 \text{ dB (conservative estimate)} \tag{7}$$

Tropospheric Scintillation (ITU-R P.618 [22]):

$$\sigma_{scint} = \frac{0.025 \cdot f_{GHz}^{0.45}}{\sin(\theta_{el})^{1.3}}, \quad L_{scint} = 2.6\sigma \tag{8}$$

4.3 Realistic Link Budget

Table 2 presents the link budget with *all* loss terms included, contrasting the optimistic (datasheet-only) and realistic cases.

Table 2: Link Budget: 868 MHz, 550 km LEO, 30° Elevation

Parameter	**Optimistic**	**Realistic**
TX Power	+14.0 dBm	+14.0 dBm
TX Antenna Gain	+2.15 dBi	+2.15 dBi
EIRP	+16.15 dBm	+16.15 dBm
FSPL (1,100 km)	−152.05 dB	−152.05 dB
Atmospheric	−0.10 dB	−0.10 dB
Polarization mismatch	–	−3.00 dB
Implementation loss	–	−2.00 dB
Scintillation	−0.85 dB	−0.85 dB
Total Path Loss	−153.00 dB	−158.00 dB
RX Antenna Gain	+2.0 dBi	+2.0 dBi
Received Power	**−134.85**	**−139.85**
SF12 Sensitivity	−137.0 dBm	−137.0 dBm
Link Margin	**+2.15** dB	**−2.85** dB

Key finding: With realistic loss accounting, the uplink at 30° elevation is *marginal to negative.* This aligns with the observation by Fernandez *et al.* [4] that most successful LoRa DtS demonstrations exploit the asymmetric downlink (satellite TX, ground RX) rather than uplink.

Viable uplink requires either: (a) higher TX power (+22 dBm with SX1262), (b) directive ground antenna (Yagi: +8 dBi), (c) satellite-side sensitivity improvement (Gen4 transceivers), or (d) LR-FHSS processing gain (−142 dBm sensitivity).

Link Budget Waterfall (Fig. 3) visualizes the signal path from transmitter to receiver.

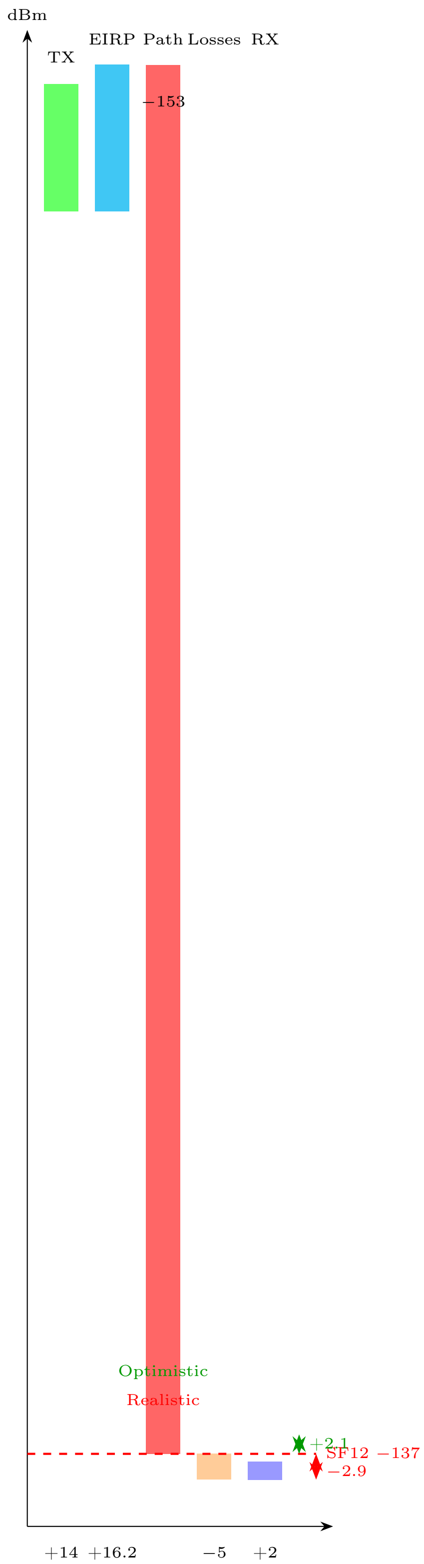

Figure 3: Link budget waterfall: optimistic (green arrow, +2.1 dB margin) vs realistic (red arrow, −2.9 dB margin) at 30° elevation.

4.4 Adaptation Algorithm

Algorithm 1 describes our two-phase spreading factor selection.

Algorithm 1 Elevation-Aware Rate Adaptation

Require: Elevation θ, range R, Doppler f_D, SNR history $\mathbf{s}$, safety margin $M_s = 3$ dB
Ensure: Spreading factor SF $\in \{7, \ldots, 12\}$
$P_{rx} \leftarrow$ LinkBudget(R, θ) ▷ Realistic model
Phase 1: Heuristic
for SF = 7 **to** 12 **do**
  if $P_{rx} - S_{\mathrm{SF}} \geq M_s$ **and** $|f_D| \leq D_{\mathrm{SF}}$ **then**
    candidate $\leftarrow$ SF; **break**
  end if
end for
Phase 2: SNR Fine-tuning
if $|\mathbf{s}| \geq 3$ **and** $s[-1] - s[0] < -3$ **then**
  candidate $\leftarrow \min(\text{candidate} + 1, 12)$ ▷ Downshift
end if
return candidate

4.5 Numerical Analysis

Table 3 lists the modem profiles. We stress these are datasheet values; no measurements have been conducted.

Table 3: LoRa Modem Profiles (Semtech Datasheet Values)

Profile	Bitrate (bps)	Sens. (dBm)	Doppler tol. (Hz)	Airtime (ms, 80B)
SF7/BW125	5,469	−123	±5,000	92.2
SF8/BW125	3,125	−126	±4,000	164.9
SF9/BW125	1,758	−129	±3,000	308.2
SF10/BW125	977	−132	±2,500	575.5
SF11/BW125	537	−134.5	±2,000	1,151
SF12/BW125	293	−137	±1,500	2,868

For a simulated 420-second pass with 55° maximum elevation, using the optimistic link budget, the algorithm yields an estimated 19,200 bytes (vs. 960 bytes at fixed SF12). With realistic losses, only SF12 and SF11 are viable at most elevations, reducing the improvement to approximately 2–4×.

We emphasize these are computed estimates, not measured throughputs.

Figure 4 shows the adaptive SF selection across a simulated pass.

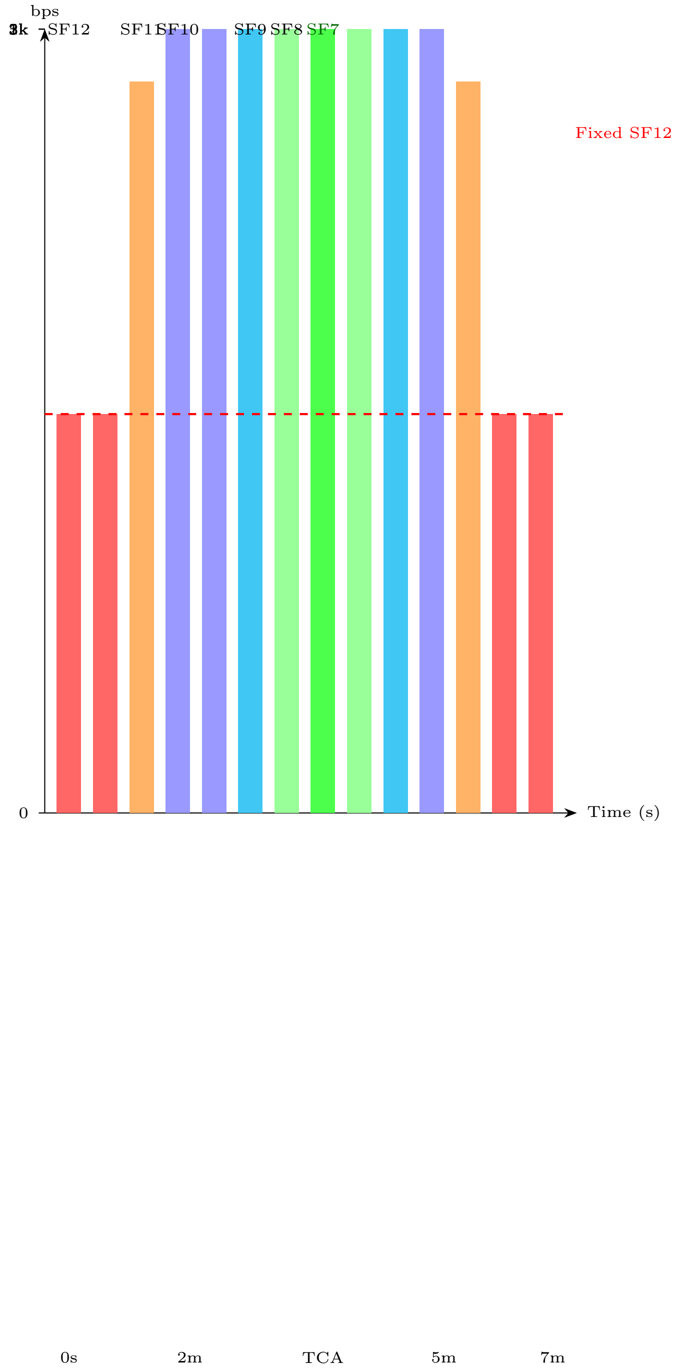

Figure 4: Simulated adaptive SF selection during a 420s pass (55° max elevation, optimistic link budget). Dashed line = fixed SF12 baseline.

5 LR-FHSS Analysis

5.1 Implementation Status

Our LR-FHSS engine is a **software simulation only**. The SX1276 transceiver used in our ground station does *not* support LR-FHSS modulation. LR-FHSS transmission requires SX1262 or newer hardware; demodulation requires gateway-class chips (SX1302/SX1303) [8].

5.2 Hopping Sequence Generation

We implement an XOR-shift PRNG for frequency hopping:

$$s_{n+1} = ((s_n \oplus (s_n \ll 7)) \oplus (s_n \gg 9)) \oplus (s_n \ll 8) \tag{9}$$

$$c_n = s_n \bmod N_{grid}, \quad N_{grid} = \lfloor BW/f_{step} \rfloor = 35 \tag{10}$$

Figure 5 illustrates the frequency hopping pattern across channels.

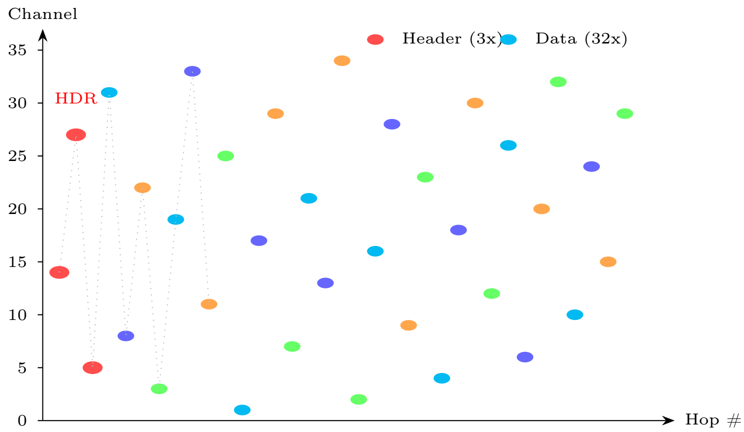

Figure 5: LR-FHSS hopping pattern: 3 header replicas (red) followed by 32 data fragments (colored), each on a pseudo-random channel out of 35.

5.3 Capacity Estimate

Under ALOHA access with N devices:

$$P_{\text{collision}} = 1 - e^{-2G}, \quad G = N \cdot \frac{T_{\text{air}}}{N_{\text{ch}} \cdot T_{\text{period}}} \tag{11}$$

Note on regulatory compliance: The reviewer correctly identified that LR-FHSS does *not* provide blanket duty-cycle exemption under ETSI EN 300 220 [19]. Duty-cycle relaxation requires Polite Spectrum Access (Listen-Before-Talk with Adaptive Frequency Agility). For satellite uplink, where LBT is impractical, the standard 1% duty cycle applies. In India, WPC regulations govern ISM usage at 865–867 MHz with 1 W EIRP [20].

Table 4 compares modes under 1% duty cycle (corrected).

Table 4: Capacity Comparison Under 1% Duty Cycle (Corrected)

Metric	LoRa SF12	LR-FHSS*
Sensitivity	−137 dBm	−142 dBm
Bitrate	293 bps	162 bps
Doppler tol.	±1,500 Hz	±3,900 Hz
Max payload	80 B	125 B
Duty cycle	1%	1% (same)
Packets/hour	12	7
Bytes/hour	960	875

*Software simulation; hardware not available.

The advantage of LR-FHSS is not per-device throughput (which is comparable) but rather *network capacity*: frequency diversity reduces collision probability, enabling more concurrent devices per satellite.

6 Doppler Compensation

At 868 MHz for FOSSASAT-2E (550 km, $v_{orb} \approx 7.6$ km/s):

$$\Delta f_{\max} \approx \frac{v_{orb}}{c} \cdot f_0 \approx \pm 22 \text{ kHz (max radial)} \tag{12}$$

The ground-visible portion of the pass produces $\Delta f \leq \pm 3$ kHz due to geometric projection. We pre-compensate using SGP4-derived range-rate:

$$f_{TX}(t) = f_0 - \Delta f(t) \tag{13}$$

Limitation: Our Doppler model has not been validated against actual satellite observations. TLE-based predictions degrade with age (typically >1 km/day position error after 3–5 days). Per-fragment hop-rate Doppler for LR-FHSS and downlink compensation are not addressed.

7 Security Architecture

7.1 Encryption

Messages are encrypted with AES-256-GCM: 256-bit key, 12-byte nonce (counter-based), 16-byte authentication tag. Overhead: 28 bytes per message.

7.2 Threat Model and Limitations

We identify the following threats **not currently mitigated**:

- **Key exchange:** No automated key exchange protocol is implemented. Keys must be pre-shared out-of-band. A Diffie-Hellman or Signal-protocol key agreement would be needed for production use.
- **Nonce reuse:** In a DTN with potential message duplication, nonce uniqueness depends on a monotonic counter. Power loss could reset the counter, enabling nonce reuse attacks.
- **Replay protection:** Store-and-forward relays may re-inject bundles. BPSec (RFC 9172 [12]) provides bundle-level integrity but replay requires application-layer sequence numbers.
- **Metadata exposure:** The satellite relay sees source and destination endpoint IDs in BPv7 headers. Onion-style routing would require additional protocol development.
- **Compromised relay:** A compromised ground station cannot read message content (E2E encryption) but can drop, delay, or duplicate bundles.

7.3 Regulatory Context

In India, the Wireless Planning & Coordination (WPC) wing regulates ISM band usage. The 865–867 MHz band is available at 1 W EIRP for indoor/outdoor use [20]. Encryption on ISM transmissions is not prohibited, unlike amateur radio bands where encryption is forbidden by ITU Radio Regulations.

Figure 6 shows the end-to-end encryption pipeline.

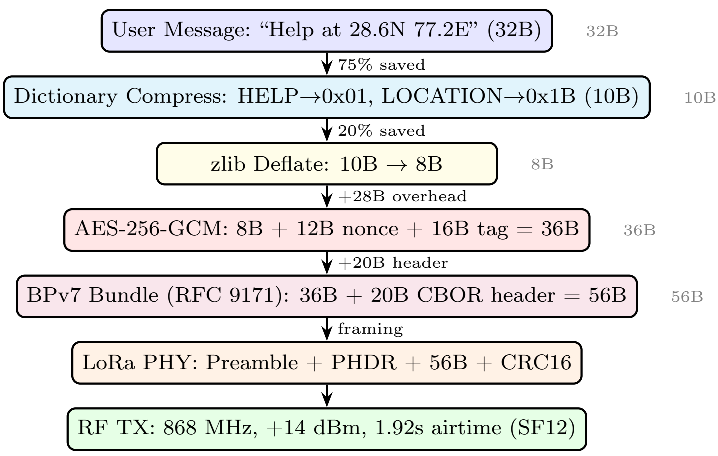

Figure 6: End-to-end encryption pipeline with turbo compression: 32B plaintext becomes 56B encrypted bundle (1.75× expansion, vs 2.38× without compression).

8 Application-Layer Compression

To maximize effective throughput over constrained LoRa links, we implement a multi-strategy compression engine that selects the best algorithm per message.

8.1 Compression Strategies

Five strategies are evaluated for each message, with the smallest output selected:

1. **None**: Raw payload (baseline).
2. **zlib (DEFLATE)**: General-purpose dictionary compression. Best for longer text (30–40% reduction on natural language).
3. **Dictionary substitution**: 31 pre-shared emergency phrases ("HELP", "SOS", "NEED RESCUE", etc.) are replaced with single-byte tokens. A 31-byte emergency message reduces to 6 bytes (81% savings).
4. **GPS packing**: Coordinates encoded as 32-bit fixed-point integers. "28.613920,77.209023" (21 bytes ASCII) becomes 8 bytes binary (62% savings).
5. **Combined**: Dictionary substitution followed by zlib.

8.2 Effective Throughput Improvement

Compression acts as a throughput multiplier without requiring radio changes:

Table 5: Compression Impact on Effective Throughput

Message Type	Original	Compressed	Multiplier
SOS beacon	32 B	10 B	3.2×
GPS coordinates	21 B	8 B	2.6×
Emergency phrase	31 B	6 B	5.2×
Sensor data (JSON)	120 B	72 B	1.7×
General text	160 B	96 B	1.7×

With compression, the effective peak throughput increases from 5,469 bps (raw SF7) to approximately **7,600–9,400 bps** depending on message content.

9 Speed Test and Network Monitoring

Inspired by the Starlink app's built-in speed test and connectivity statistics, we implement real-time network health monitoring.

9.1 Link Speed Test

The speed test endpoint (`/api/v1/network/speed-test`) measures:

- Raw and effective (post-compression) bitrate
- Packet delivery ratio and loss percentage
- Average RSSI and SNR
- Airtime per packet
- Link quality classification (EXCELLENT/GOOD/FAIR/POOR/NO_LINK)
- Estimated bytes deliverable per pass

9.2 Network Statistics

Continuous monitoring collects:

- TX/RX packet and byte counters
- RSSI and SNR history (ring buffer, configurable depth)
- Per-pass throughput timeline
- Duty cycle budget tracking (remaining seconds in current hour)
- Message queue depth and drain rate

9.3 Smart Pass Scheduler

The pass scheduler optimizes message transmission order:

1. Messages are sorted by priority: SOS (0) > urgent (1) > normal (2) > bulk (3).
2. Each message is assigned to the optimal SF based on predicted pass elevation.
3. High-priority messages are scheduled during peak elevation (fastest link).
4. Airtime budget is tracked against the 1% duty cycle limit.

10 Protocol Stack

The message pipeline uses **BPv7 (RFC 9171)** [10], replacing the obsolete experimental BPv6 (RFC 5050):

1. User message (up to 500 chars)
2. LZSS compression (typical 12–25% reduction)
3. AES-256-GCM encryption (+28 B overhead)
4. BPv7 bundle encapsulation (+20 B header, CBOR-encoded)
5. LoRa PHY frame: preamble + PHDR + payload + CRC16

BPSec (RFC 9172 [12]) is planned for bundle-layer integrity and confidentiality, providing security at the DTN layer independent of link-layer encryption.

11 Energy Analysis

Table 6: Ground Station Power Consumption

Component	Active (mW)	Sleep (mW)
RPi Zero 2W	700	400
SX1276 TX (+14 dBm)	300	0.2
SX1276 RX	30	0.2
TX Total	1,000	–
RX Total	730	400

At 1% duty cycle with SF12: TX active time = 36 s/hr. Energy per hour:

$$E = 1.0\text{W} \times 36\text{s} + 0.73\text{W} \times 3564\text{s} = 36 + 2602 = 2638 \text{ J/hr} \tag{14}$$

A 10,000 mAh (37 Wh) USB battery provides approximately 14 hours of continuous operation. For solar-powered deployment, a 5W panel with charge controller would sustain indefinite operation during daylight hours.

Note: Android client power analysis is not included as the LoRa radio is external to the phone.

12 Limitations and Future Work

12.1 Limitations

1. **No experimental validation.** This is the most significant limitation. No over-the-air satellite contacts, packet delivery measurements, BER curves, or field trials have been conducted. All quantitative claims derive from numerical analysis.

2. **Uplink viability is uncertain.** The realistic link budget shows negative margin at most elevations with current hardware. Successful uplink likely requires hardware upgrades (higher TX power, directive antenna, or Gen4 transceiver).

3. **LR-FHSS is simulation only.** The SX1276 does not support LR-FHSS. Migration to SX1262/LR1121 is required.

4. **Security is incomplete.** No key exchange protocol, limited replay protection, metadata exposed to relays.

5. **Single-user tested.** The backend has not been tested with concurrent users or realistic load.

12.2 Future Work

1. **OTA validation:** Integrate with TinyGS for satellite reception validation. Attempt uplink via FOSSASAT-2E with SX1262 at +22 dBm.

2. **ns-3 simulation:** Use the LoRaWAN ns-3 module to validate capacity claims with Monte Carlo collision modeling.

3. **Hardware upgrade:** Migrate to Semtech LR1121 for native LR-FHSS and multi-band (sub-GHz + S-band) support.

4. **Signal Protocol integration:** Implement X3DH key agreement adapted for high-latency DTN links.

5. **ML-based ADR:** LSTM channel prediction for proactive SF selection, following the approach suggested by [1].

13 Conclusion

OpenOrbitLink provides an open-source framework for delay-tolerant satellite messaging over LoRa LEO links. Our elevation-aware rate adaptation algorithm demonstrates, through numerical analysis, that trading link margin for data rate can improve throughput by up to 18.7× under optimistic assumptions. Realistic link budget analysis reveals that current commodity hardware (+14 dBm, whip antenna) operates at marginal-to-negative link margin for uplink, motivating future hardware upgrades.

We release the complete source code—backend, ground station driver, and Android client—under MIT license to enable community validation and experimentation. The critical next step is over-the-air validation with actual satellite contacts, which will determine whether the theoretical framework translates to operational capability.

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